

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

May 2012

EE205 Electrical Machines I

Date: 15.05.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Develop a relationship between energy and co energy in a magnetic field system. [07]

Also give a diagram of flow of energy in an electromechanical energy conversion device via a coupling field.

Q - 2 (a) Derive the emf equation in a dc machine from fundamentals and explain why lap winding is preferred for low voltage high current machines. [07]

(b) A 250 V, 14.9 kW, 8 pole dc motor has single turn coils. The armature is wave wound with 94 commutator segments. If the brushes are shifted by 2 commutator segments at full load, calculate (i) total armature reaction ampere turns (ii) demagnetizing ampere turns (iii) cross magnetizing ampere turns. Assume motor efficiency to be 80 %.

OR

Q - 2 (a) In a long shunt compound generator, the terminal voltage is 230 V when it delivers 150 A. Determine: (i) induced emf (ii) total power generated by armature. The shunt field, series field, diverter and armature resistances are 92, 0.015, 0.03 and 0.032 ohm respectively. [07]

(b) How the voltage is build up in a dc shunt generator? Sketch the magnetization characteristics of a dc shunt generator. From the characteristics give the expression for the critical speed and critical resistance. [07]

Q - 3 (a) Explain the methods of speed control of dc shunt motors. [07]

(b) Explain why starter is required for starting a dc motor. Explain with neat sketch the working of a 3 point starter of a dc shunt motor. [07]

OR

Q - 3 (a) Explain the different types of losses taking place in a dc machines. Derive the condition for maximum efficiency. [07]

(b) Draw and explain the external characteristics of dc shunt, series and compound generator. [07]

SECTION – II

Q - 4 Draw and explain the complete phasor diagram for a transformer when the load power factor is lagging. [07]

Q - 5 (a) Explain the different types of losses taking place in a transformer. On what factors do they depend? Explain the steps taken to minimize these losses. [07]

(b) A 10 kVA, 200/400 V, 50 Hz single phase transformer gave the following test results. [07]

Open circuit test(hv winding open): 200 V, 1.3 A, 120 W

Short circuit test (lv winding short circuited): 22 V, 30 A, 200 W.

Find the parameters of equivalent circuit as referred to lv winding.

OR

Q - 5 (a) Explain an auto transformer? State its merits and demerits over the two winding transformer. [07]

(b) Find all day efficiency of a transformer having maximum efficiency of 98% at 15 kVA at unity power factor and loaded as follows: [07]

12 hours: 02 kW at 0.5 power factor lag;

06 hours: 12 kW at 0.8 power factor lag;

06 hours: at no load.

Q - 6 (a) Draw the physical connection and phasor diagram of the following transformer connections (i) D z 6 (ii) Y d 1 (iii) Y z 11 [07]

(b) Explain the necessary conditions required for the parallel operation of transformer. [07]

OR

Q - 6 (a) Discuss the different methods of cooling of transformer. [07]

(b) Describe the back to back test for the separation of losses in two identical transformers. [07]
